

L2A™

Q2. (a) Discuss the biological and cultural factors that led to evolution of modern homo sapiens.
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Modern Homo Sapiens are the result of series of ~~evolutionary~~ factors from Australopithecus to Homo erectus to present day humans.

Biological factors

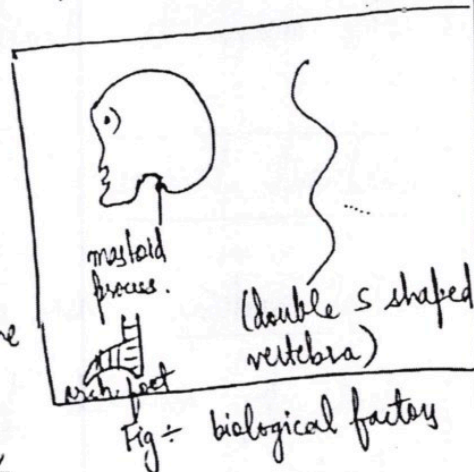
- 1) Increased size and complexity of brain
 - cranial capacity increased from 400cc (A. africanus) to around 1400cc
 - increased development of neuron cells in the brain
 - highly developed nervous system giving rise to increased intelligent ability.

2) Erect posture and Bipedalism

- development of mastoid process and linear aspera

- forward and hollowed placement of foramen magnum $\Rightarrow$ shifts center of gravity $\rightarrow$ towards spine

- development of extensor muscles helped in erect posture.



3) Remodelling of face and teeth $\Rightarrow$ decreased prognathism & parabolic dental arcade.

4) Stereoscopic vision due to bipedalism & erect posture

5) Reduced size of digestive system led to increased energy extraction for other purposes.

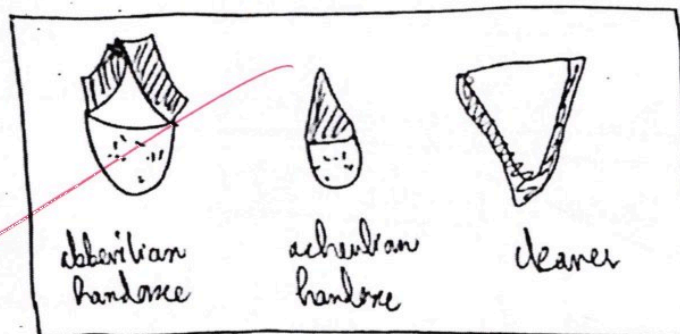
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6) Development of hyoid bone which enabled the man to develop speech

Cultural Factors

1) Tool making : use of Acheulian & Abbevilian tools by enabled faster work and efficient hunting.



2) Hunting and gathering led to increased communication & coordination $\Rightarrow$ also ensured food security.

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- 3) Use of Fire by Homo erectus increased the reach to colder areas and enabled them to eat cooked food
- 4) Emergence of social organisation like marriage, political organisation which further led to evolution.
- 5) Development of speech and language allowed them to communicate and coordinate properly.
- 6) Neolithic food production $\Rightarrow$ ensured surplus and sedentary economy.

Hence, a host of factors enabled the evolution of Homo sapiens which started nearly 2.5 million years ago.

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sup for your points by any
found evidences